2024 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT) | 979-8-3315-3313-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/3ICT64318.2024.10824248

AI Phishing Detection Framework for Businesses with Limited Resources

Rhythm Jaiswal
CSE Dept.
Ajay Kumar Garg Engg. College
Ghaziabad, India
rhythmjaiswal116@gmail.com

Marshal R
ICERT
New Delhi, India
marshalrgj@gmail.com

Kirti Pal Singh
ICERT
New Delhi, India
kirti.pal.91@gmail.com

V. Venkateswara Rao
ICERT, MeitY
New Delhi, India
venkat4u.rao@gmail.com

***Abstract*— Phishing remains the predominant entry point in most of the cyber-attacks. Artificial Intelligence based Phishing attacks are difficult to detect and evade the traditional phishing detection techniques. Detection of such phishing emails requires sophisticated tools. However, small businesses and startups face challenges to procure such sophisticated tools for monitoring phishing emails. In this work, an Artificial Intelligence phishing detection framework is proposed to detect Artificial Intelligence based phishing emails including Quishing attacks. The robust framework can be used by organizations having limited resources. The proposed methodology is validated by applying the framework on Artificial Intelligence generated phishing emails. The robustness of the proposed framework can be used to enhance the cybersecurity posture of organizations against phishing attacks.**

Keywords*— *Artificial Intelligence (AI), Attacks, Cyber, Phishing, Quishing, Threats

I. Introduction

Phishing is the widely used social engineering tactic to get access to systems by threat actors. Threat actors have changed the phishing tactics and campaigns over the years. They are still used as an effective mechanism and attack vector ranging from small scale to large scale sophisticated attacks. With human beings being the weakest link in the system, phishing campaigns are carried out in different forms through emails, voice, short messages and chats in social media platforms. Threat actors use email phishing largely to target organizations and get access to the network of the victim organization. Several reports indicate the rise in phishing and phishing being the key entry point for cyber-attacks, cyber frauds and cybercrimes [1, 2].

Threat actors have changed their tactics with time and with the rise in Artificial Intelligence (AI), threat actors are using AI to send crafted phishing emails that are hard to detect using the conventional phishing detection mechanisms used by organizations [3, 4]. Threat actors are leveraging the potential of AI for malicious purposes and mitigating the threats has become harder with the rise in AI attacks. Small businesses and start-ups have limited financial resources and they face more challenges in allocating the limited funds to Cyber security. AI phishing detection tools are available. However, affordability of such tools increases the financial burden of organizations. To address this challenge, in this work, an AI phishing detection framework is developed. The proposed framework is also validated by applying the framework on phishing emails generated using AI. The proposed framework can be used as a baseline framework by small and medium scale organizations for detection AI generated Phishing emails. It can help them to cover the common and simple attacks that are targeted at them by using AI. The proposed framework may be further enhanced by using Machine learning algorithms and by combining the proposed framework along with the conventional phishing detection mechanisms [5-8].

The rest of the article is designed as follows. Section II focusses on the common trends deployed in AI phishing attacks. Section III explores the commonly used AI phishing detection tools. Section IV elaborates the proposed framework and methodology for AI phishing detection along with the results obtained. Section V discusses the work done and the future work that can be done.

II. AI Phishing Attacks

A. *Email Phishing*

Phishing remains the widely used social engineering tactic. Email phishing remains one of the most commonly used entry vectors for criminals. The threat actors exploit the human weakness and change their tricks continuously and try to get access to networks of the victim organization [9-12]. The common email phishing tricks used by threat actors is presented in Fig. 1.

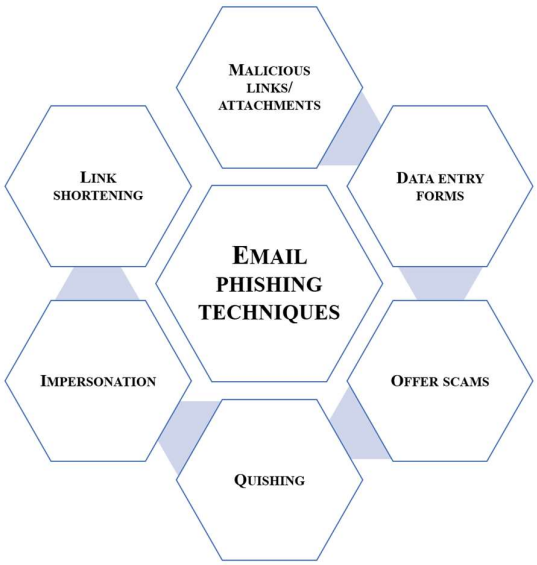

Fig. 1. Email phishing tricks

Authorized licensed use limited to: UNIVERSITI TEKNOLOGI MARA. Downloaded on April 28,2025 at 04:31:59 UTC from IEEE Xplore. Restrictions apply.

Threat actors send emails with malicious links or attachments and clicking on such links or attachments drops a malware into the user system and slowly the threat actors get access to the entire network through lateral movement. In some cases, the threat actors send emails with lucrative offers or impersonate themselves as senior officials and make the user to commit a mistake by clicking on any link or attachment or forces them to share sensitive data. Advanced threat actors use genuine online forms and collect sensitive information and evade the phishing detection techniques used at mail servers and the security solutions deployed at endpoints to identify malicious activities [13, 14]. They also use QR codes as they are not detected by email servers.

B. *AI in Phishing*

With the rise in AI, threat actors have started to use AI for malicious activities including generation of phishing emails. AI enabled attacks are more sophisticated phishing attempts [15,16]. Small-scale industries and individuals are particularly vulnerable to phishing emails due to weaker security measures, lack of awareness and fewer resources to invest in robust cybersecurity measures compared to large corporations. With AI, fraudsters are able to scale their attacks and cause long term damages. Leveraging AI, threat actors are able to send polished, personalized and legitimate-looking emails. The use of generative models to tweak and improve messages has enabled scammers to scale exponentially to exploit specific vulnerabilities. The Natural Language Generation (NLG) skills of AI is used to impersonate individuals or organisations that are hard to detect by the existing phishing detection techniques. The key attributes of AI that are exploited for phishing email generation is presented in Fig. 2.

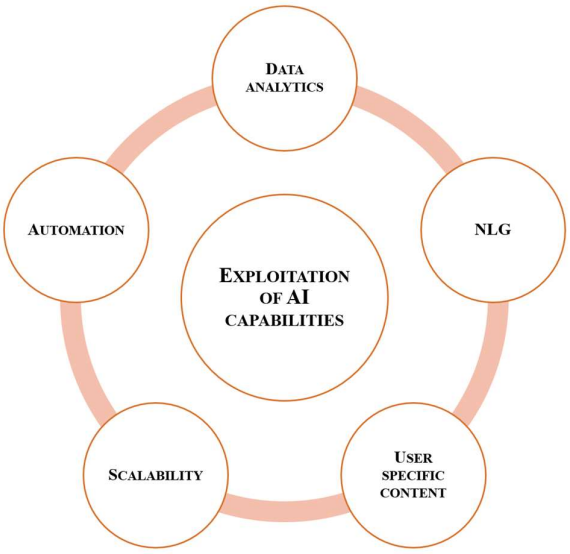

Fig. 2. AI abilities exploited for phishing

The capability of AI to crawl huge amount of data and collect information about each user helps the threat actors to filter the profiles of the target users. The NLG capability and the ability to write customized texts that look more authentic and relevance based on the user profile is widely exploited to generate emails that are too hard to detect as a phishing email as the content seems to be more genuine. The ability of AI to be scaled largely with limited resources and the automation opportunities that are offered by AI makes AI based phishing more affordable and attractive for threat actors.

III. AI PHISHING DETECTION TOOLS

To detect malicious phishing emails, several Advanced Threat Protection (ATP) techniques are proposed in the literature [17, 18]. Some of the commonly and widely used tools for AI generated Phishing email are discussed in this section [19, 20].

A. *Microsoft ATP*

The solution from Microsoft uses AI and ML technologies to detect and block phishing emails. It analyses email content, sender reputation, and user behaviour to identify potential threats and prevent them from reaching the inbox.

B. *PhishMe*

PhishMe, now known as Cofense, provides a range of anti-phishing solutions that combine AI with human intelligence. Their tools simulate phishing attacks to educate employees and use machine learning to analyse email content and metadata to detect phishing attempts.

C. *Barracuda Sentinel*

Barracuda Sentinel is another effective security solution used in detecting phishing emails. By using modern AI and ML technologies, the tool tries to provide a comprehensive protection to organizations against phishing attacks. They use advanced techniques that detect the phishing attempts and helps to prevent the infiltration of phishing emails into the organization. Barracuda Sentinel, with its intelligent approach guarantees secure communication routes and uncompromised security channels.

D. *Agari*

Agari provides tools and services that can be utilized to spot and protect against phishing attempts. They analyze the legitimacy and genuineness of the emails and provide real-time feeds on phishing frauds and blocks the malicious emails at the perimeter of the enterprises. The solution also helps to provide protection against the leakage of sensitive data from the network and help to strengthen the proactive defense posture of the organization.

E. *Challenges for small scale businesses*

Although these AI tools provide excellent services, most small-scale industries are incapable to utilize them during the early stages of establishments due to financial constraints. This is widely exploited by threat actors and they target small businesses and startups that lack sophisticated tools and skilled manpower. The lack of awareness on AI based phishing attacks among the users is also an increasing concern as AI based attacks are hard to differentiate from genuine emails and it requires more technical knowledge to distinguish the phishing emails in most cases as conventional phishing detection techniques are not that much effective in AI phishing emails.

Authorized licensed use limited to: UNIVERSITI TEKNOLOGI MARA. Downloaded on April 28,2025 at 04:31:59 UTC from IEEE Xplore. Restrictions apply.

IV. PROPOSED METHODOLOGY

Security devices deployed in an environment regularly look for malicious behaviour and try to detect the intrusion attempts made by the threat actors and alert the defenders. However, AI phishing emails use modern techniques in addition to the conventional techniques. Hence, it becomes complicated and difficult to detect such malicious emails. In this work, the rule-based analysis is leveraged to detect and mitigate the AI phishing techniques by designing a framework to identify phishing emails and warn the users of suspicious activity.

A diverse dataset of phishing and legitimate emails was considered for the study including different AI phishing emails that are available in open source [21-24]. Based on the patterns and techniques observed in such emails, a framework was developed that covers and identifies the common techniques observed in AI phishing emails. The proposed framework focuses on the key aspects observed in phishing emails is presented in Fig. 3. The framework is robust and can be implemented by small businesses. It requires minimal manpower and can detect common phishing emails and can enhance the cyber resilience of small businesses against AI phishing attacks.

A. *AI generated Text Detection*

Often, we have seen phishing emails contain AI generated texts curated especially to target customers or recipients into giving sensitive information. In this work, Sapling.ai detector is used to analysis the textual data present in the email. It prints the score after analysing the message and warns out the presence of AI generated content in the email. If the score is less than the threshold value then the text is likely to be human generated. Based on the score, the alert is provided for potential phishing email as presented in Fig. 4.

B. *Zero-Font Detection:*

Zero-font detection is a method commonly used by the attackers to bypass detection systems. They often employ this technique to hide malicious content from the recipient while ensuring it remains visible to search engines and text-based filtering systems. The proposed model identifies these phishing attempts in the .eml file sent to the recipient by scanning the email content for text styled with a font size of zero. It is a small but crucial detection point which can be helpful in ensuring the legitimacy of the email. The results for zero font detection are presented in Fig.5.

C. *Attachment Analysis:*

Most emails contain some form of file attachments for the purpose of communication. This makes it easier for the phishers to often use large or suspicious attachments to deliver malwares such as viruses, worms, trojan horse etc to the targeted system. Most of these malware files are sent in the form of executable file attachments which are triggered by clicking on the file. In this work, the email is analysed for detection of any such attachments present in the email using the Virus Total API by generating SHA-256 hash values for the file. By scrutinizing these attachments, including their file types and sizes, it is possible to detect and mitigate hidden threats. Analysing email attachments involves inspecting not only their presence but also their size and content. The results obtained through VirusTotal API for the hash values generated for the attachments are presented in Fig. 6.

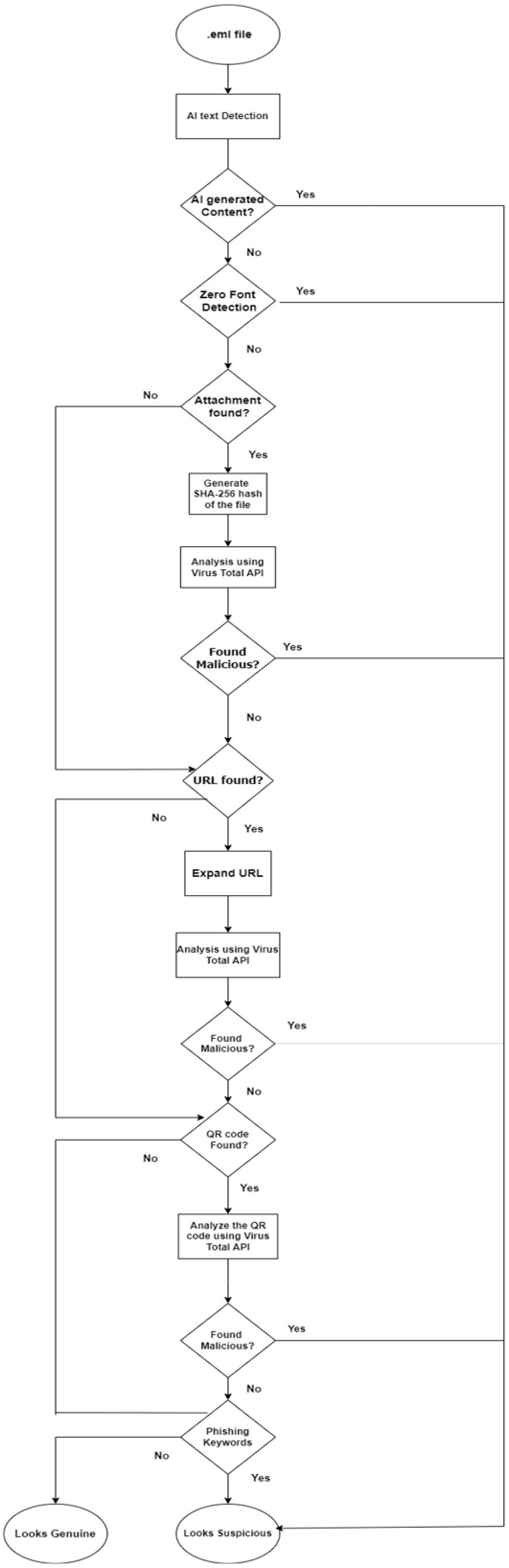

Fig. 3. Proposed framework

Authorized licensed use limited to: UNIVERSITI TEKNOLOGI MARA. Downloaded on April 28,2025 at 04:31:59 UTC from IEEE Xplore. Restrictions apply.

```
Rounded Score: 1.242527595e-06
The text is likely to be human-written.

{'score': 1.2425275949246029e-06,
 'sentence_scores': [{'score': 0.077572299090226, 'sentence': 'Dear Suresh,'},
                     {'score': 0.5142261721622947,
                      'sentence': 'I am speaking on behalf of Bank of Baroda, '
                                  'Janakpuri.'},
                     {'score': 0.005702861868660647,
                      'sentence': 'I came across your profile on LinkedIn.'},
                     {'score': 0.9600086426642495,
                      'sentence': 'As a financial expert in this sector for '
                                  'the past 10 years, I am highly impressed by '
                                  'your skill set and experience.'},
                     {'score': 0.009519895495864739,
                      'sentence': 'I wish to offer you the position of Branch '
                                  'Manager in our Janakpuri branch and become '
                                  'a part of our family.'},
                     {'score': 0.9823155387222652,
                      'sentence': 'Kindly fill in your details in the form '
                                  'given below.'},
                     {'score': 0.05344784474362396,
                      'sentence': 'Best Regards,'},
                     {'score': 0.0, 'sentence': 'HR department'}],
 'text': 'Dear Suresh,\n'
         '\n'
         'I am speaking on behalf of Bank of Baroda, Janakpuri. I came across '
         'your profile on LinkedIn. As a financial expert in this sector for '
         'the past 10 years, I am highly impressed by your skill set and '
         'experience. I wish to offer you the position of Branch Manager in '
```

Fig. 4. AI generated Text Analysis

```
Sender: phisher@example.com
Subject: Urgent: Verify Your Account Information

Links:
http://bit.ly/verify-account
   - This link may be associated with a URL shortener (potential phishing)
http://bit.ly/verify-account
   - This link may be associated with a URL shortener (potential phishing)

Attachments:

Phishing Keywords Detection:
   - Potential phishing keywords detected in the email content.

Zero Font Detection:
   - Zero font size text detected in the email (potential obfuscation).
```

Fig. 5. Zero font detection

D. *Phishing Language Detection:*

In general, phishing emails contain specific key terms and phrases to create a sense of panic or urgency and use emotional weakness of users to reveal sensitive information. The social engineering methods use techniques to trick the victims by exploiting psychological factors.

Therefore, it is important to analyse the language of the email to identify a phishing attempt. In this work, detection of common phishing terms, such as “urgent” “verify” “update” “password” “login” “immediately” is carried out to detect fraudulent terms that are commonly used in a phishing attack. Fig.7 shows the detection of the phishing keywords detected by the proposed Framework

E. *Macilious URL Detection:*

Phishers frequently use URL shortening services such as bitly, tinyURL, and rebrandly to conceal malicious links and redirect the recipient to an illegitimate site. Detecting and expanding these shortened URLs to reveal their final destination is a critical step in identifying potential phishing attacks. The given framework uses the Virus Total API key to inspect the links present in the email. The data fetched from the API will help in uncovering hidden threats masked behind seemingly innocuous short URLs. The URLs are validated by API integration with Virus total database and the links are checked for flagging by AV vendors in Virus Total. Based on the response the malicious email alert can be raised. The results achieved by API integration with Virustotal is presented in Fig. 8. QR code analysis is also carried out in the proposed framework to identify malicious URLs.

F. *Pros and Cons of the proposed framework*

The proposed framework is developed using rule based analysis for detection of phishing attempts. The simplicity of these rules allow for quick deployment and immediate results, essential for timely threat mitigation. This approach offers high predictability and control over the detection process. Security teams can easily understand, monitor, and modify the rules, ensuring transparency in the detection mechanism. The proposed model requires minimal computational resources compared to advanced machine learning models. This efficiency makes it an attractive option for organizations with limited resources.

However, hard-coded rules are static in nature and struggle to adapt to evolving tactics, rendering them less effective over time. Rule based learning are often prone to vulnerabilities and bypassing techniques. Security teams need to constantly monitor and revise these rules, which can be a significant burden. Small businesses can use the proposed model as a guideline and can start deploying small ML models curated to their needs and can improve their protection against phishing attacks.

```
File Hash (SHA-256): 7b42b24371d4a3231f7935d6099abe24681baa862c653f491

VirusTotal Analysis Results:
+----------------------+------------------+----------+-----------+
| Engine               | Category         | Result   | Method    |
+======================+==================+==========+===========+
| Bkav                 | undetected       |          | blacklist |
+----------------------+------------------+----------+-----------+
| Lionic               | undetected       |          | blacklist |
+----------------------+------------------+----------+-----------+
| tehtris              | undetected       |          | blacklist |
+----------------------+------------------+----------+-----------+
| ClamAV               | undetected       |          | blacklist |
+----------------------+------------------+----------+-----------+
| CMC                  | undetected       |          | blacklist |
+----------------------+------------------+----------+-----------+
| CAT-QuickHeal        | undetected       |          | blacklist |
+----------------------+------------------+----------+-----------+
| Skyhigh              | undetected       |          | blacklist |
+----------------------+------------------+----------+-----------+
| ALYac                | undetected       |          | blacklist |
+----------------------+------------------+----------+-----------+
| Cylance              | undetected       |          | blacklist |
+----------------------+------------------+----------+-----------+
| Zillya               | undetected       |          | blacklist |
+----------------------+------------------+----------+-----------+
| Sangfor              | undetected       |          | blacklist |
+----------------------+------------------+----------+-----------+
| K7AntiVirus          | undetected       |          | blacklist |
+----------------------+------------------+----------+-----------+
| K7GW                 | undetected       |          | blacklist |
```

Fig. 6. Results for Attachment Analysis

Authorized licensed use limited to: UNIVERSITI TEKNOLOGI MARA. Downloaded on April 28,2025 at 04:31:59 UTC from IEEE Xplore. Restrictions apply.

```
Sender: "dot&keycares@gmail.com" <dot&keycares@gmail.com>
Subject: Fill in you bank details for purchase

Links:
https://phishingsite=2Ecom/verify-accou=

Attachments:
Name: 1.jpg | Size: 1342208 bytes

Phishing Keywords Detection:
   - Potential phishing keywords detected in the email conte
     - verify
     - account
     - immediately
```

Fig. 7. Phishing keywords detection

```
input_url = "http://ramajudicial.info"
get_url_analysis(input_url)
```

Acronis	harmless	clean
Abusix	harmless	clean
ADMINUSLabs	harmless	clean
Lionic	harmless	clean
Criminal IP	harmless	clean
AILabs (MONITORAPP)	harmless	clean
AlienVault	harmless	clean
alphaMountain.ai	harmless	clean
AlphaSOC	undetected	unrated
Antiy-AVL	harmless	clean
ArcSight Threat Intelligence	undetected	unrated
AutoShun	undetected	unrated
benkow.cc	harmless	clean
Bfore.Ai PreCrime	malicious	malicious
BitDefender	harmless	clean
Bkav	undetected	unrated
BlockList	harmless	clean
Blueliv	harmless	clean
Certego	harmless	clean

Fig. 8. URL Analysis

V. CONCLUSION AND FUTURE WORK

The use of AI in phishing is a huge challenge in cyber security as the existing detection mechanisms are not effective enough to detect such threats. Phishing being the key entry vector in most of the threats, AI phishing emails pose a greater challenge to the Cyber defenders. However, by deploying effective security measures, it is possible to defend against these evolving threats. In this work, a robust framework is proposed for detecting AI based phishing emails. The framework incorporates a set of heuristic-based detection points, including the use of zero font, attachment analysis, and other suspicious characteristics. The results obtained indicate the effectiveness and reliability of these heuristics in distinguishing phishing emails from legitimate ones. The framework can be used by smaller organizations with limited resources and also by start-ups as a baseline tool in their cyber security preparedness against AI phishing attempts. As rule based techniques have their own limitations, addition of ML based approaches will further strengthen the cyber resilience against phishing attempts. By continually updating and enhancing the proposed framework with new detection points and ML techniques, small organizations can easily enhance their protection against AI phishing threats.

VI. REFERENCES

[1] H. Abroshan et al., "Phishing Happens Beyond Technology: The Effects of Human Behaviors and Demographics on Each Step of a Phishing Process," in *IEEE Access*, vol. 9, pp. 44928-49, 2021.

[2] F. Castaño et al., "PhiKitA: Phishing Kit Attacks Dataset for Phishing Websites Identification," in *IEEE Access*, vol. 11, pp. 40779-89, 2023.

[3] A. Chien and P. Khethavath, "Email Feature Classification and Analysis of Phishing Email Detection Using ML Techniques," CSDE, Nadi, Fiji, 2023, pp. 1-8.

[4] Basit Aet al., "A comprehensive survey of AI-enabled phishing attacks detection techniques". *Telecommun Syst.* vol. 76, no. 1, pp. 139-154, 2021.

[5] N. Z. Gorment et al., "ML Algorithm for Malware Detection: Taxonomy, Current Challenges, and Future Directions," in *IEEE Access*, vol. 11, pp. 141045-89, 2023.

[6] R. Kaur, D. Gabrijelčič, T. Klobučar, "Artificial intelligence for cybersecurity: Literature review and future research directions," *Inf. Fusion*, vol. 97, 101804, 2023.

[7] S. P. Ripa et al., "The Emergence Threat of Phishing Attack and The Detection Techniques Using ML Models," ACMI, Rajshahi, Bangladesh, 2021, pp. 1-6.

[8] V. Patil et al., "Detection and Prevention of Phishing Websites Using Machine Learning Approach," ICCUBEA, Pune, India, 2018, pp. 1-5.

[9] J. Hajny et al., "Framework, Tools and Good Practices for Cybersecurity Curricula," in *IEEE Access*, vol. 9, pp. 94723-47, 2021.

[10] A. Karim et al., "Phishing Detection System Through Hybrid Machine Learning Based on URL," in *IEEE Access*, vol. 11, pp. 36805-22, 2023.

[11] R. Zieni et al.,"Phishing or Not Phishing? A Survey on the Detection of Phishing Websites," in *IEEE Access*, vol. 11, pp. 18499-519, 2023.

[12] P. Bountakas et al.,"HELPHED: Hybrid Ensemble Learning PHishing Email Detection", *J. Netw. Comput. Appl.*, vol. 210, 103545, 2023.

[13] Michael Cobb, "16 common types of cyberattacks and how to prevent them" 2024 [Online] Available: https://www.techtarget.com/searchsecurity/tip/6-common-types-of-cyber-attacks-and-how-to-prevent-them (Accessed on 24 July 2024)

[14] Security, "Coronavirus-related spear phishing attacks see 667% increase" [Online] Available: https://www.securitymagazine.com/articles/92157-coronavirus-related-spear-phishing-attacks-see-667-increase-in-march-2020. (Accessed on 24 July 2024)

[15] Tom Burt, "Microsoft Digital Defense Report, 2023." [Online] Available: https://www.microsoft.com/en-us/security/security-insider/microsoft-digital-defense-report-2023 (Accessed on 24 July 2024)

[16] APWG, "Phishing Activity Trends Report 1st Quarter 2024 [Online] Available: https://docs.apwg.org/reports/apwg_trends_report_q1_2024.pdf (Accessed on 24 July 2024)

[17] S. Salloum et al.,"Phishing Email Detection Using NLP Techniques: A Literature Survey," *Procedia Comput. Sci.*, Vol. 189, pp. 19-28, 2021.

[18] S. Atawneh et al.,"Phishing Email Detection Model Using DL", *Electronics*, vol. 12, no. 20, pp. 4261, 2023.

[19] Microsoft, "Anti-phishing protection in Microsoft 365" [Online] Available at: https://learn.microsoft.com/en-us/defender-office-365/anti-phishing-protection-about (accessed on July 2024)

[20] Ushma Singh, "Top 15 Best Anti-Phishing Tools and Services: A Comprehensive Guide" [Online] Available at: https://www.tikaj.com/blog/anti-phishing-tools-and-services/ (Accessed on 24 July 2024).

[21] S. Tiwari, "Phishing Dataset for Machine Learning," [Online]. Available: https://www.kaggle.com/datasets/shashwatwork/phishing-dataset-for-machine-learning. [Accessed: Oct. 19, 2024].

[22] A. Prasad and S. Chandra, "PhiUSIIL: A diverse security profile empowered phishing URL detection framework based on similarity index and incremental learning," *Computers & Security*, vol. 136, pp. 103545, 2024, doi: 10.1016/j.cose.2023.103545.

[23] S. Marchal, J. Francois, R. State, and T. Engel, "PhishStorm: Detecting Phishing with Streaming Analytics," *IEEE Transactions on Network and Service Management*, vol. 11, no. 4, pp. 458-471, 2014.

Authorized licensed use limited to: UNIVERSITI TEKNOLOGI MARA. Downloaded on April 28,2025 at 04:31:59 UTC from IEEE Xplore. Restrictions apply.

[24] T. M. A. Tamal, M. K. Islam, T. Bhuiyan, and A. Sattar, "Dataset of suspicious phishing URL detection," *Frontiers in Computer Science*, vol. 6, p. 1308634, 2024, doi: 10.3389/fcomp.2024.1308634.

Authorized licensed use limited to: UNIVERSITI TEKNOLOGI MARA. Downloaded on April 28,2025 at 04:31:59 UTC from IEEE Xplore. Restrictions apply.